

Ignatius Ryan Pranantyo, PhD

Natural Hazard Scientist | Tsunami – Storm Surge – Catastrophe Modelling

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PROFESSIONAL SUMMARY

I build hazard models that turn physical uncertainty into actionable risk intelligence. With a decade spanning government disaster risk programmes, research, and catastrophe model development at a model vendor firm, I understand what the science needs to deliver — not just in publications, but in production-ready model frameworks, commercial client outputs, and cross-team coordination. Specialising in coastal flood from tsunamis and storm surges across the APAC region; skilled in deterministic, stochastic, and probabilistic approaches at regional to asset scale.

WORK EXPERIENCE

Research Fellow

Earth Observatory of Singapore, Nanyang Technological University, Singapore | January 2025 - present

- Developed a modular, two-stage PTHA framework to make large-scale tsunami inundation modelling tractable along the 1,000 km southern coast of Java — coupling offshore (JAGURS) and onshore (SFINCS) simulations across 4,400 unique events at 30 m inundation resolution, built entirely on open-source software and data; framework, scripts, and hazard database released openly via GitHub and NTU Research Data Repository.
- Delivered tsunami scenario design and inundation simulations (30 m resolution) along the DKI Jakarta coastline for Maipark's R&D initiative exploring tsunami as a new insurable peril; findings co-presented at AOGS 2025.
- Provided tsunami scenario design and hazard analysis for 2 collaborative projects: input preparation for a port risk-assessment GUI tool (Tanjung Garem Port, Banten) and model setup and analysis advisory for a regional hazard assessment (northern Sulawesi); both manuscripts under review (Geoscience Letters and Marine Geophysical Research, 2026).
- Presented at the International Tsunami Symposium 2025 (poster) and UN Ocean Decade Tsunami Programme 2025; upcoming oral presentation at EGU General Assembly, May 2026.

Catastrophe Model Developer

Reask, United Kingdom | March 2022 - January 2025

- Led development of stochastic storm surge coastal flood databases for 3 regions (the Philippines, Hong Kong–Macau–Pearl River Delta, and Europe) commissioned by AXA Paris; established the modelling framework, simulated 20,000+ historical and synthetic tropical cyclone events at up to 30 m resolution for key areas, and delivered all outputs on time with full calibration-validation and technical documentation — subsequently adopted into AXA's internal risk workflows.
- Built the offshore storm surge component of a new US TC-CAT product developed with an external partner — a multi-team effort spanning wind/rain, coastal, and pluvial/fluvial flood disciplines; simulated 100,000+ offshore storm surge events and supported framework design, dataset preparation, and model calibration-validation.
- Designed a proof-of-concept near-real-time storm surge forecasting system to support Reask's TC forecasting product; automated model setup and simulation pipeline, reducing operational turnaround from days to hours.
- Developed the storm surge simulation framework and scientific documentation for a parametric insurance proposal for Pacific Nations; proposal subsequently awarded and currently in implementation.

Career Break

July 2021 – March 2022

Planned relocation to Japan for a research position was disrupted by COVID-19 travel restrictions. Continued scientific writing; completed and published source reconstruction study of the 1969 Western Sulawesi earthquake and tsunami (Pure and Applied Geophysics, 2022).

Postdoctoral Research Fellow

Brunel University London, United Kingdom | January 2020 - July 2021

- Contributed as researcher to the Royal Society-funded project “Building Earthquake and Tsunami Resilience in East Indonesia”, coordinating across Brunel University London, LIPI (now BRIN), and Indonesian partner institutions.
- Led and published 2 first-author studies on complex tsunami hazards in eastern Indonesia – covering seismic and non-seismic sources (Geoscience Letters, 2021) and source reconstruction of the 1969 Western Sulawesi earthquake and tsunami (Pure and Applied Geophysics, 2022).
- Co-investigated 2 rapid-response studies on the 2019 Molucca Sea (Mw7.2) and 2020 Aegean Sea (Mw7.0) earthquake and tsunamis; both published in peer-reviewed journals (Seismological Research Letters and Pure and Applied Geophysics, 2021).

Tsunami Modeller

Self-employed, Indonesia | August 2019 - January 2020

- Completed PhD thesis revisions while finalising research outputs from doctoral work at the Australian National University.
- Developed 25-m resolution deterministic tsunami inundation models for a mining area in collaboration with Badan Geologi (Geological Agency of Indonesia), including full technical documentation.
- Delivered JAGURS tsunami modelling training for Badan Geologi staff, building in-house modelling capacity for independent hazard assessment.

PhD Researcher

Australian National University, Australia | June 2015 - July 2019

- Pioneered systematic tsunami hazard studies for eastern Indonesia – a historically understudied region – through source identification and reconstruction of 3 historical events (1674 Ambon, 1852 Banda Sea, and 1992 Flores); contributing foundational literature that has since inspired subsequent studies in the region.
- Published 2 first-author papers in Pure and Applied Geophysics and co-authored 1 paper in Nature Geoscience; applied multiple inversion methodologies to identify compound earthquake-landslide tsunami mechanisms.
- Delivered ANUGA tsunami modelling training for Indonesian government collaborators in partnership with Geoscience Australia, building in-house hazard assessment capacity across multiple institutions.

Jr. Geophysicist

Australia - Indonesia Facility for Disaster Reduction (AIFDR), Indonesia | October 2012 - May 2015

A bilateral programme between AusAID/DFAT (Australia) and BNPB (Indonesia)

- Served as lead contributor (2nd author) to Indonesia's first national-scale probabilistic tsunami hazard assessment, supporting the government's disaster risk reduction programme; work formed the basis of MSc thesis and subsequent peer-reviewed publications.
- Co-authored a heterogeneous slip model study for tsunami hazard assessment in the Mentawai Islands (Geological Society, London, Special Publications, 2017), building on research initiated during this role.
- Delivered 10+ ANUGA tsunami modelling workshops for Indonesian government institutions, producing 4 official maps in collaboration with Badan Geologi and establishing an in-house modelling capacity that continues to the present day.

EDUCATION

PhD in Seismology & Mathematical Geophysics

Australian National University, Australia | June 2015 - July 2019

Tsunami hazard in eastern Indonesia: Source identification and reconstruction for historical case studies

Australia Awards Scholarship, DFAT (Australian Government)

MSc in Earth Sciences

Bandung Institute of Technology, Indonesia | July 2011 - October 2012

Probabilistic tsunami hazard assessment for the Banda Sea, eastern Indonesia

Accelerated programme (commenced MSc coursework in the final year of BSc)

BSc in Oceanography

Bandung Institute of Technology, Indonesia | July 2007 - June 2011

The 2010 Mentawai Islands tsunami modelling

Merit-based scholarship, ITB (awarded to the top 10 students per semester)

TECHNICAL SKILLS

Tsunami and storm surge modelling	JAGURS, SCHISM, SFINCS, ANUGA, COMCOT, OCSMesh
Earthquake modelling	OpenQuake
Programming	Python (pandas, NumPy, xarray, SciPy, matplotlib), Fortran (working knowledge)
Geospatial analysis	QGIS, Generic Mapping Tools (GMT)
Computing	HPC (on-premise & cloud-based), Linux, Windows
Version control & environment	Git, conda

Additional Information

Memberships	European Geoscience Union (EGU); Global Tsunami Model (GTM) working group
Languages	English (professional); Bahasa Indonesia (native)
Publications	12 peer-reviewed articles h-index: 12 Google Scholar: link
Work Rights	Singapore Employment Pass holder; open to relocation with visa sponsorship
References	Available upon request